



Entrega T9

Métodos Matemáticos Para La Física

1. Hallar los vectores velocidad y aceleración y la ecuación de la recta tangente para la siguiente curva:

a) $\sigma(t) = (\sin 3t, \cos 3t, 2t^{3/2}), t = 1$

b) $\sigma(t) = (t \sin t, t \cos t, \sqrt{3}t), t = 0$

a) $\sigma(t) = \sin(3t)\vec{i} + \cos(3t)\vec{j} + 2t^{3/2}\vec{k}$

$$v(t) = \frac{d\sigma(t)}{dt} = (3\cos(3t), -3\sin(3t), 3t^{1/2})$$

$$v(1) = (3\cos(3), -3\sin(3), 3)$$

$$a(t) = \frac{dv(t)}{dt} = (-9\sin(3t), -9\cos(3t), \frac{3}{2}t^{-1/2})$$

$$a(1) = (-9\sin(3), -9\cos(3), \frac{3}{2})$$

Recta Tangente en $t=1 \rightarrow$ Recta con pendiente $v(1)$ que pasa por $\sigma(1)$

Ecuación de la recta: $y = mx + c$

$$r(s) = v(1)s + \sigma(1) = (3\cos(3), -3\sin(3), 3)s + (\sin(3), \cos(3), 2)$$

$$\sigma(1) = (\sin(3), \cos(3), 2)$$

$$r(s) = (3\cos(3)s + \sin(3), -3\sin(3)s + \cos(3), 3s + 2)$$

$$b) \sigma(t) = (t \operatorname{sen}(t), t \cos(t), \sqrt{3} t), t=0$$

$$\sigma(0) = (0, 0, 0)$$

$$v(t) = (\operatorname{sen}(t) + t \cos(t), \cos(t) - t \operatorname{sen}(t), \sqrt{3})$$

$$v(0) = (0, 1, \sqrt{3})$$

$$a(t) = (2 \cos(t) - t \operatorname{sen}(t), -2 \operatorname{sen}(t) - t \cos(t), 0)$$

$$a(0) = (2, 0, 0)$$

Recta tangente en $t=0$:

$$r(s) = v(0)s + \overset{(0,0,0)}{\sigma(0)} = (0, 1, \sqrt{3})s + (0, 0, 0) = (0, s, \sqrt{3}s)$$

2. Probar la siguiente regla para la trayectoria diferenciables en $\mathbb{R}^2$ y $\mathbb{R}^3$

$$\frac{d}{dt}[\sigma(t) \times \rho(t)] = \frac{d\sigma}{dt} \times \rho(t) + \sigma(t) \times \frac{d\rho}{dt}$$

Aplicamos Definición de la Derivado: $f'(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t+\Delta t) - f(t)}{\Delta t}$

$$\frac{d[\sigma(t) \times \rho(t)]}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\sigma(t+\Delta t) \cdot \rho(t+\Delta t) - \sigma(t) \cdot \rho(t)}{\Delta t}$$

Sumamos y Restamos por $\sigma(t+\Delta t) \cdot \rho(t)$ para poder agrupar términos

$$= \lim_{\Delta t \rightarrow 0} \frac{\sigma(t+\Delta t) \cdot \rho(t+\Delta t) - \sigma(t) \cdot \rho(t) + \sigma(t+\Delta t) \cdot \rho(t) - \sigma(t+\Delta t) \cdot \rho(t)}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{\sigma(t+\Delta t) [\rho(t+\Delta t) - \rho(t)] + \rho(t) [\sigma(t+\Delta t) - \sigma(t)]}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \sigma(t+\Delta t) \frac{\rho(t+\Delta t) - \rho(t)}{\Delta t} + \rho(t) \frac{\sigma(t+\Delta t) - \sigma(t)}{\Delta t}$$

$$\text{(cuando } t \rightarrow 0: \sigma(t+\Delta t) \approx \sigma(t), \frac{\rho(t+\Delta t) - \rho(t)}{\Delta t} = \frac{d\rho(t)}{dt}$$

$$\frac{d[\sigma(t) \times \rho(t)]}{dt} = \sigma(t) \cdot \frac{d\rho(t)}{dt} + \rho(t) \frac{d\sigma(t)}{dt}$$

3. Calcular la longitud de arco de la curva en el intervalo indicado

a) $\sigma(t) = (\sin 3t, \cos 3t, 2t^{3/2})$ en $[0, 1]$

b) $\sigma(t) = ti + tj + \frac{2}{3}t^{3/2}k$, en $[a, b]$

c) $\sigma(t) = (t, t \sin t, t \cos t)$ entre $(0, 0, 0)$ y $(\pi, 0, -\pi)$

$$L(c) = \int_{t_0}^{t_1} \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt$$

a) $\sigma(t) = (\sin 3t, \cos 3t, 2t^{3/2}) \quad t \in [0, 1]$

$$x(t) = \sin 3t \quad x'(t) = 3 \cos(3t)$$

$$y(t) = \cos 3t \quad y'(t) = -3 \sin(3t)$$

$$z(t) = 2t^{3/2} \quad z'(t) = 3t^{1/2}$$

$$L(c) = \int_0^1 \sqrt{9 \cos^2(3t) + 9 \sin^2(3t) + 9t} = \int_0^1 \sqrt{9 + 9t} = 3 \int_0^1 \sqrt{1+t}$$

$$L(c) = 3 \left[\frac{2}{3} (1+t)^{3/2} \right]_0^1 = 3 \left(\frac{2}{3} \cdot 2^{3/2} - \frac{2}{3} \cdot 1^{3/2} \right) = 2 (2^{3/2} - 1) u = 3,657 u$$

$$b) \sigma(t) = (t, t, \frac{2}{3}t^{3/2}) \quad t \in [a, b]$$

$$\begin{aligned} x(t) &= t & x'(t) &= 1 \\ y(t) &= t & y'(t) &= 1 \\ z(t) &= \frac{2}{3}t^{3/2} & z'(t) &= t^{1/2} \end{aligned}$$

$$L(c) = \int_a^b \sqrt{2+t} \, dt = \left[\frac{2}{3}(2+t)^{3/2} \right]_a^b = \frac{2}{3}[(2+b)^{3/2} - (2+a)^{3/2}]$$

$$c) \sigma(t) = (t, t \sin(t), t \cos(t)) \quad \alpha(t) \in [(0,0,0), (\pi, 0, -\pi)]$$

$$\left. \begin{aligned} \sigma(t) &= (0, 0, 0) \rightarrow t = 0 \\ \sigma(t) &= (\pi, 0, -\pi) \rightarrow t = \pi \end{aligned} \right\} t \in [0, \pi]$$

$$\begin{aligned} x(t) &= t & x'(t) &= 1 \\ y(t) &= t \sin(t) & y'(t) &= \sin(t) + t \cos(t) \\ z(t) &= t \cos(t) & z'(t) &= \cos(t) - t \sin(t) \end{aligned}$$

$$d(c) = \int_0^\pi \sqrt{1 + (\sin(t) + t \cos(t))^2 + (\cos(t) - t \sin(t))^2} \, dt$$

$$(\sin(t) + t \cos(t))^2 = \sin^2(t) + 2t \sin(t) \cos(t) + t^2 \sin^2(t)$$

$$(\cos(t) - t \sin(t))^2 = \cos^2(t) - 2t \sin(t) \cos(t) + t^2 \sin^2(t)$$

$$d(c) = \int_0^\pi \sqrt{2+t^2} \, dt = \left[\frac{1}{2}t \sqrt{2+t^2} + \ln \left| \frac{\sqrt{2}}{2}t + \sqrt{\frac{1}{2}(2+t^2)} \right| \right]_0^\pi$$

$$L(c) = \frac{\pi \sqrt{\pi^2+2}}{2} + \ln \left(\frac{\sqrt{2}}{2} + \sqrt{\frac{\pi^2}{2}+1} \right) u = 6,95 \, u$$

4. Hallar los puntos críticos y su naturaleza (máximo, mínimo o punto silla)

a) $f(x, y) = x^2 + y^2 + 2xy$

b) $f(x, y) = e^{1+x^2-y^2}$

c) $f(x, y) = \cos(x^2 + y^2)$

d) $f(x, y) = y + x \sin y$

e) $f(x, y) = \log(2 + \sin xy)$

a) $f(x, y) = x^2 + y^2 + 2xy$

$$f_x = 2x + 2y \rightarrow f_x = 0 \rightarrow 2x + 2y = 0 \rightarrow x = -y$$

$$f_y = 2y + 2x \quad f_y = 0 \rightarrow 2y + 2x = 0 \rightarrow y = -x$$

Todo punto del tipo $(x, -x)$ es un punto crítico

$$f_{xx} = 2 \quad d = f_{xx}(x, -x) \cdot f_{yy}(x, -x) - (f_{xy}(x, -x))^2 = 0$$

$$f_{yy} = 2$$

$$f_{xy} = 2$$

No podemos aplicar esto

$$f(x, y) = (x^2 + y^2)^2 \rightarrow f(x, -x) = 4x^2 \quad f(x, -x) > 0 \quad \forall x \in \mathbb{R}$$

Todos los puntos $(x, -x)$ son mínimos

$$b) f(x,y) = e^{1+x^2-y^2}$$

$$\begin{aligned} f_x &= 2x e^{1+x^2-y^2} \rightarrow f_x = 0 \rightarrow x=0 \\ f_y &= -2y e^{1+x^2-y^2} \rightarrow f_y = 0 \rightarrow y=0 \end{aligned}$$

Punto crítico en el $(0,0)$

$$\begin{aligned} f_{xx} &= (4x^2 + 2) e^{1+x^2-y^2} \\ f_{yy} &= (4y^2 - 2) e^{1+x^2-y^2} \end{aligned}$$

$$f_{xy} = -4xy e^{1+x^2-y^2}$$

$$d = f_{xx}(0,0) f_{yy}(0,0) - (f_{xy}(0,0))^2 = \begin{vmatrix} f_{xx}(0,0) & f_{xy}(0,0) \\ f_{yx}(0,0) & f_{yy}(0,0) \end{vmatrix}$$

$$d = 2e \cdot -2e - 0^2 = -4e$$

$d < 0 \rightarrow$ Punto silla en $(0,0,e)$

$$c) f(x,y) = \cos(x^2+y^2)$$

$$\left. \begin{aligned} f_x &= -2x \sin(x^2+y^2) \\ f_y &= -2y \sin(x^2+y^2) \end{aligned} \right\} x^2+y^2 = n\pi \quad \forall n \in \mathbb{Z}$$

$$\begin{aligned} f_{xx} &= -2 \sin(x^2+y^2) - 4x^2 \cos(x^2+y^2) \\ f_{yy} &= -2 \sin(x^2+y^2) - 4y^2 \cos(x^2+y^2) \end{aligned}$$

$$f_{xy} = -4xy \cos(x^2+y^2)$$

$$\text{Caso } (x,y) = (0,0) \rightarrow d = f_{xx}(0,0) \cdot f_{yy}(0,0) - (f_{xy}(0,0))^2 = 0 \cdot 0 - 0 = 0$$

Analizamos $f(0,0) = 1 \rightarrow (0,0)$ es un máximo ya que el coseno está limitado entre $[-1, 1]$

caso $x^2 + y^2 = nn$

$$\begin{aligned} f_{xx} &= -2 \cancel{\sin(n\pi)} - 4x^2 \cancel{\cos(n\pi)}^{(-1)^n} = -4x^2 (-1)^n \\ f_{yy} &= -4y^2 (-1)^n \\ f_{xy} &= -4xy (-1)^n \end{aligned}$$

$$d = f_{xx} \cdot f_{yy} - (f_{xy})^2 = -4x^2 \cdot -4y^2 - 16x^2y^2 = 0 \quad \text{no sirve el analisis}$$

$$f(x,y) = \cos(n\pi)$$

- Si n es impar, es un minimo ya que $\cos(n\pi) = -1$
- Si n es par, es un maximo ya que $\cos(2n\pi) = 1$

d) $f(x,y) = y + x \sin y$

$$f_x = \sin y \rightarrow f_x = 0 \rightarrow \sin y = 0 \rightarrow y = k\pi$$

$$f_y = 1 + x \cos y \rightarrow f_y = 0 \rightarrow x \cos y = -1 \quad x = (-1)^{k+1} = -(-1)^k$$

Puntos criticos del tipo $(-1)^{k+1}, k\pi$

$$f_{xx} = 0$$

$$f_{yy} = -x \sin y$$

$$f_{xy} = \cos y$$

$$\begin{aligned} d &= f_{xx} \cdot f_{yy} - (f_{xy})^2 = 0 \cdot -x \sin y - \cos^2 y \\ &= -\cos^2 y \end{aligned}$$

$$d < 0 \quad \forall y \in \mathbb{R}$$

$$f((-1)^k, k\pi) = k\pi + \cancel{(-1)^{k+1} \sin k\pi}^{=0} = k\pi$$

Los puntos tipo $(-1)^{k+1}, k\pi, k\pi$ son puntos silla

$$e) \ln(2 + \sin xy) \quad \frac{v}{v} = \frac{vdu - u dv}{v^2}$$

$$f_x = \frac{y \cos xy}{2 + \sin xy} \rightarrow f_x = 0 \quad 2 + \sin xy \neq 0 \quad y \cos xy = 0 \begin{cases} y=0 \\ \cos xy=0 \end{cases}$$

$$\cos xy = 0 \quad x \cdot y = \frac{n\pi}{2} \quad \forall n \in \mathbb{N} / n \text{ es impar}$$

$$f_y = \frac{x \cos xy}{2 + \sin xy} \rightarrow f_y = 0 \rightarrow x \cos xy = 0 \begin{cases} x=0 \\ \cos xy=0 \end{cases} \rightarrow x \cdot y = \frac{n\pi}{2} \quad \forall n \in \mathbb{N} / n \text{ es impar}$$

$$\text{Puntos Críticos: } (0,0), \left(\frac{n\pi}{2}, \frac{n\pi}{2}\right)$$

$$f_{xx} = \frac{-(2 + \sin xy)(y^2 \sin xy) - y^2 \cos^2 xy}{(2 + \sin xy)^2}$$

$$f_{yy} = \frac{-(2 + \sin xy)(x^2 \sin xy) - x^2 \cos^2 xy}{(2 + \sin xy)^2}$$

$$f_{xy} = \frac{(2 + \sin xy)(\cos xy - xy \sin(xy)) - xy \cos^2(xy)}{(2 + \sin xy)^2}$$

$$d = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

$$d = -\left(\frac{(-1)^n}{2}\right)^2 = -\frac{1}{4}$$

$$\text{caso } (0,0):$$

$$d < 0 \rightarrow (0,0, \ln(2)) \text{ es un punto de silla}$$

$$f_{xx} = 0$$

$$f_{yy} = 0$$

$$f_{xy} = \frac{2 \cdot (-1)^n}{2^2} = \frac{(-1)^n}{2}$$

Caso: $x \cdot y = \frac{n^2}{2} = \frac{1}{2} + n^2 \quad \forall n \in \mathbb{N}$

$$f_{xx} = \frac{-(2+(-1)^k)(y^2(-1)^k)}{4}$$

$$f_{yy} = \frac{-(2+(-1)^k)(x^2(-1)^k)}{4}$$

$$f_{xy} = \frac{(2+(-1)^k)(-xy(-1)^k)}{4}$$

$d = 0 \rightarrow$ No podemos aplicar

$$f(x,y) = \ln(2 + (-1)^k)$$

Si k par $\rightarrow f(x,y) = \ln(3) = \text{Valor Máximo}$

Si k impar $f(x,y) = \ln(1) = 0 = \text{Valor Mínimo}$

En Resumen:

Punto de silla en $(0,0, \ln(2))$,

Máximo cuando $x \cdot y = \frac{1}{2} + k^2$ con $k \in \mathbb{Z} / k = 2n$ (par)

Mínimo cuando $x \cdot y = \frac{1}{2} + k^2$ con $k \in \mathbb{Z} / k = 2n-1$ (impar)

5. Hallar los extremos de f sujetos a las restricciones indicadas

a) $f(x, y, z) = x - y + z, x^2 + y^2 + z^2 = 2$

b) $f(x, y) = x, x^2 + 2y^2 = 3$

c) $f: \mathbf{R}^2 \rightarrow \mathbf{R}, (x, y) \mapsto x^2 - y^2, S = \{(x, \cos x) | x \in \mathbf{R}\}$

a) $f(x, y, z) = x - y + z$ Restricción: $x^2 + y^2 + z^2 = 2$

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\vec{\nabla} f = (1, -1, 1) \quad \vec{\nabla} g = (2x, 2y, 2z)$$

$$1 = 2x\lambda, \quad -1 = 2y\lambda, \quad 1 = 2z\lambda$$

$$x = \frac{1}{2\lambda}, \quad y = -\frac{1}{2\lambda}, \quad z = \frac{1}{2\lambda}$$

$$\left(\frac{1}{2\lambda}\right)^2 + \left(-\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 2 \rightarrow \frac{3}{4\lambda^2} = 2 \rightarrow 3 = 8\lambda^2$$

$$\lambda^2 = \frac{3}{8} \rightarrow \lambda = \pm \sqrt{\frac{3}{8}} = \frac{\sqrt{3}}{2\sqrt{2}} = \frac{2\sqrt{6}}{8} = \pm \frac{\sqrt{6}}{4}$$

$$\begin{aligned} \text{Caso } \lambda = \frac{\sqrt{6}}{4} &\rightarrow 1 = \frac{\sqrt{6}}{2} x \rightarrow x = \frac{2\sqrt{6}}{6} & -1 = \frac{\sqrt{6}}{2} y \rightarrow y = -\frac{2\sqrt{6}}{6} \\ & 1 = \frac{\sqrt{6}}{2} z \rightarrow z = \frac{2\sqrt{6}}{6} & \text{Punto } \left(\frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}\right) \end{aligned}$$

$$\text{Caso } \lambda = -\frac{\sqrt{6}}{4} \rightarrow \text{Punto } \left(-\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}\right)$$

evaluamos f :

$$f\left(\frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}\right) = \frac{2\sqrt{6}}{6} + \frac{2\sqrt{6}}{6} + \frac{2\sqrt{6}}{6} = \sqrt{6}$$

$$f\left(-\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}\right) = -\sqrt{6}$$

$$P_1: \left(\frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}\right) \Rightarrow \text{Maximo}$$

$$P_2: \left(-\frac{2\sqrt{6}}{6}, \frac{2\sqrt{6}}{6}, -\frac{2\sqrt{6}}{6}\right) \Rightarrow \text{Minimo}$$

b) $f(x, y) = x$ Restricción: $x^2 + 2y^2 = 3$

$$\vec{\nabla} f = \lambda \nabla g$$

$$\vec{\nabla} f = (1, 0) \quad \vec{\nabla} g = (2x, 4y)$$

$$1 = 2x\lambda \rightarrow x = \frac{1}{2\lambda} \quad 4y\lambda = 0 \rightarrow y = 0$$

$$\left(\frac{1}{2\lambda}\right)^2 = 3 \rightarrow \frac{1}{4\lambda^2} = 3 \rightarrow 12\lambda^2 = 1 \rightarrow \lambda^2 = \frac{1}{12} \rightarrow \lambda = \pm \frac{\sqrt{3}}{6}$$

$$\text{Caso } \lambda = \frac{\sqrt{3}}{6} \rightarrow \frac{\sqrt{3}}{3} x = 1 \rightarrow x = \sqrt{3} \quad P_1(\sqrt{3}, 0)$$

$$\text{Caso } \lambda = -\frac{\sqrt{3}}{6} \rightarrow x = -\sqrt{3} \quad P_2(-\sqrt{3}, 0)$$

$$f(\sqrt{3}, 0) = \sqrt{3} \rightarrow \text{Maximo}$$

$$f(-\sqrt{3}, 0) = -\sqrt{3} \rightarrow \text{Minimo}$$

$$c) f: \mathbb{R}^2 \rightarrow \mathbb{R} (x, y) \rightarrow x^2 - y^2 \quad \text{Restricción: } y = \cos x \quad f(x, y) = x^2 - \cos^2(x)$$

$$f'(x) = 2x + \sin(2x) \quad f'(x) = 0 \rightarrow x = 0$$

$$f''(x) = 2 + 2\cos(2x) \quad f''(0) = 2 + 2 = 4 \quad 4 > 0$$

Hay un mínimo en $x = 0$

$$y = \cos(x) \rightarrow y = 0$$

$$f(0, 1) = 0^2 - 1^2 = -1 \rightarrow \text{Mínimo Global en el } (0, 1)$$

6. Diseñar una lata cilíndrica con tapa que contenga 1 litro de agua, usando la mínima cantidad de metal.

$$V = \pi r^2 h = 1000 \text{ cm}^3 \rightarrow h = \frac{1000}{\pi r^2}$$

$$\text{Área Superficie} = 2\pi r^2 + 2\pi r h \rightarrow \text{Área} = 2\pi r^2 + 2\pi r \cdot \frac{1000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}$$

$$\text{Para minimizar la cantidad de metal: } A'(r) = 4\pi r - \frac{2000}{r^2} = 0$$

$$4\pi r = \frac{2000}{r^2} \rightarrow 4\pi r^3 = 2000 \rightarrow r^3 = \frac{500}{\pi} \rightarrow r = 5,42 \text{ cm}$$

$$h = \frac{1000}{\pi r^2} = 10,42 \text{ cm}$$

Lata con base de radio 5,42 cm y Altura 10,42 cm